

Association between sperm quality, oxidative stress, and seminal antioxidant activity.



OBJECTIVES: To determine seminal antioxidant capacity, oxidative stress markers, and their association with semen quality as oxidative stress is considered to be a major etiological factor in male infertility. SUBJECTS AND METHODS: Semen samples were obtained from 138 men and categorized on the basis of sperm count, motility, and morphology. Seminal oxidative and antioxidant markers are as follows: lipid peroxidation (LPO), protein carbonyls (PC), superoxide dismutase (SOD), catalase (CAT), thiols, and ascorbic acid were determined. RESULTS: Sperm count significantly correlated positively with progressive sperm motility and normal morphology. Sperm count and normal morphology showed significant negative correlation with LPO and PC. Sperm count and progressive motility showed significant positive relationship with SOD. The SOD, CAT, and thiols positively whereas LPO and PC negatively associated with elevated sperm count. CONCLUSION: Insufficient antioxidant enzymes and increased oxidative stress may attribute to the risk of declining semen quality and hence protective role for antioxidant enzymes against the oxidative damage cannot be ruled out. Copyright © 2010 The Canadian Society of Clinical Chemists. Published by Elsevier Inc. All rights reserved.